



THE COPPER CONUNDRUM

65%

Import dependence is the highest in copper, among base metals

EVERY ELECTRIC VEHICLE rolling out in India, every new transmission line strung across to carry solar power to homes, every artillery shell forged for the armed forces—all of it depends on a handful of base metals. For decades, India prided itself on producing these at home. Copper for motors and grids, aluminium for aircraft frames to power cables. However, on each of these fronts, India's dependence on imported metals is rising.

As recently as 2021-22, India ran a healthy surplus of over \$12 billion in base metal trade. The country exported more aluminium, copper, zinc, and iron ore than it imported. This position fit comfortably with India's self-image as a resource-rich nation with a maturing industrial base. By 2022-23, that picture changed. The surplus didn't shrink gradually; it flipped into a deficit. Today, India imports nearly \$52 billion worth of base metals annually, a figure now almost 43% higher than what it exports.

For an economy that has staked its future on manufacturing-led growth through Make in India, production-linked incentive (PLI) schemes, and an ambitious defence indigenisation drive, it means that even as India builds more electric vehicles (EVs), lays more grid infrastructure, and manufactures more defence equipment, the metallic backbone of all that activity is increasingly arriving by ship rather than coming out of an Indian mine or smelter.

METAL BY METAL

When the entire import bill is broken down, a clear pattern emerges. Iron and steel lead the list at \$15.8 billion, followed closely by copper and aluminium. Together, these three metals account for over 85% of all of India's base metal imports. The

trade problem is heavily concentrated in three metals, each with its own underlying story: copper's supply shock, iron ore's quality mismatch, and aluminium's own cost and capacity pressures, with domestic smelting capacity struggling to keep pace with demand from packaging, construction, and EVs and electronics, where lightweighting is a priority.

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Until just a few years ago, India was a net exporter of refined copper. The country had the smelting capacity, the processing infrastructure, and the export volumes to be a net positive contributor to global copper trade. Today, that position has completely reversed. India's copper import bill has crossed \$12.8 billion, or over ₹1 trillion during 2025-26 alone, making it the second-largest base metal imported after iron and steel.

What changed? Two forces—the demand side and the supply side collided. On the demand side, India's economy is undergoing an electrification wave that is fundamentally copper-intensive. EVs use three to four times more copper than conventional vehicles, owing to their electric motors, battery systems, and wiring.

As EV adoption accelerates, driven by policy incentives, falling battery costs, and growing consumer acceptance, the copper requirement per vehicle sold has effectively multiplied.

It isn't just EVs. Even with India's renewable energy push, from solar parks to wind farms to battery storage systems, copper demand is increasing. Solar panels need copper for their internal wiring and junction boxes. Wind turbines use copper in their generators and transformers. Battery storage systems are also copper-heavy by design.

On the supply side, Vedanta's Sterlite copper smelter in Thoothukudi,